

COUNTING TECHNIQUES AND PROBABILITY

COUNTING TECHNIQUES QUESTIONS SOLVED (2.21-2.48)

Q21) Registrants are offered 6 tours on each of 3 days. In how many ways can a person arrange to go on a sightseeing tour?

Basic Counting: 3 days and each day we can choose 6 tours so $3 \times 6 = 18$.

Q22) In medical study, patients are classified in 8 ways according to their blood group and also according to whether their BP is low, moderate, high. In how many ways a patient can be classified?

8 blood type ways and 3 BP ways so total ways of classification = $8 \times 3 = 24$.

Q23) If an experiment consists of throwing a dice and then drawing a letter at random. How many points are in Sample Space?

6 choices of dice and for each one of 6. We have 26 choices of letters so $6 \times 26 = 156$.

Q24) Students are classified as freshman, sophomore, junior and senior, and also according to their gender (M or F). Find total possible classifications.

4 year wise and 2 gender wise classifications so total = $4 \times 2 = 8$.

Q25) A brand of shoe comes in 5 different styles and each style is available in 4 distinct colors. How many different combinations of shoes?

$5 \times 4 = 20$.

Q26) Study shows that 7 health rules can extend a man's life by 11 years on average and a woman's life by 7 years. In how many ways can a person adopt these 5 rules if:

1) He is currently violating all rules?

Since a person only needs to select 5 out of the 7 rules. We will do 7 combinations 5 (as order doesn't matter). Hence $7C5 = 21$.

2) He is following 2 out of these 7 rules?

Now only 3 rules are left to follow and since 2 are selected, we have 5 rules left from to select. So we do $5C3=10$.

Q27) Home buyers are given a choice of 4 designs, 3 heating systems, a garage or carport, and a patio or porch. How many different plans are available to the buyer?

Since 4 designs, 3 heating, choice from garage or carport (so 2 options) and a choice from patio or porch (so 2 options). Total different plans = $4 \times 3 \times 2 \times 2 = 48$.

Q28) A medicine can be purchased from 5 different manufacturers in liquid, tablet, capsule which comes in regular and extra strength category. How many different ways do we have to purchase the medicine?

Since 5 manufacturers, 3 forms and 2 categories: Total ways = $5 \times 3 \times 2 = 30$.

Q29) 3 racecars are tested using 5 different gasoline at 7 test sites. If 2 drivers are used in the study and test runs are made once under each distinct set of conditions. How many test runs are needed?

$3 \times 5 \times 7 \times 2 = 210$.

Q30) In how many different ways can a T/F test with 9 Questions be answered?

Since each Question has 2 options: $2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 2^9 = 512$.

Q31) License Plate contains Letters RLH followed by 3 digits with first one being 5. How many digits total plates of this combination can exist if each digit is distinct.

Since 1 number is selected (5). We have 9 numbers left for 2nd digit and 8 for 3rd digit so $8 \times 9 = 72$. Therefore 72 plates of this combination can exist.

Q32)

a) In how many ways can 6 people be lined up to get on a bus?

$6! = 720$.

b) If 3 specific persons among 6 refuse to follow each other. How many ways are possible now?

If we let these 3 follow each other as 1 unit, then we have ABC, p4, p5, p6 hence $4! = 24$. In the ABC unit the 3 people can be arranged in $3! = 6$ ways. Therefore total ways of this scenario = $24 \times 6 = 144$ ways when those 3 follow each other. Since we don't want this: We do $720 - 144 = 576$.

Q33) MCQ test of 5 questions each with 4 choices and 1 correct answer exist.

a) In how many ways can a student check off one answer to each question?

For each question: 4 options and we have 5 questions: So $4^5 = 1024$ ways.

- b) In how many ways can a student check off one answer to each question and get all the answers wrong?

Since 3 answers are wrong: $3^5=243$.

Q34) How many distinct permutations can be made from the letters of the word COLUMNS. How many of these permutations start with letter M?

7 total words and no repeat: $7!=5040$ ways. Now since M is fixed we have 6 different letters combinations left so $6!=720$.

Q35) A contractor wants to build 9 houses with each having a different design. In how many ways can he place the houses on a street if 6 lots are on one side and 3 lots on the other side of the street?

Since all objects are distinct and can be placed on all different locations. We will simply do $9!$.

Q36) How many 3 digit numbers can be formed from 0,1,2,3,4,5,6.

- a) If each digit is only used once

$7! = 5040$.

- b) How many of these are odd numbers

$5040/2 = 2520$.

- c) How many are greater than 330

If 1st position has 4,5,6. Then we can simply do $3 \times 5 \times 6 = 90$. If 1st position has 3 then we can do $1 \times 3 \times 5 = 15$. Therefore total numbers = 105.

Q37) In how many ways can 4 boys and 5 girls sit in a row in an alternate arrangement?

First seat is filled by any of the 5 girls and second by any of the 4 boys. Then 3rd by any of 4 girls and 4th by any of 3 boys.

So $5 \times 4 \times 4 \times 3 \times 3 \times 2 \times 2 \times 1 \times 1 = 2880$.

Q38) 4 married couples have bought 8 seats in the same row for a concert. In how many different ways can they be seated?

- a) With no restrictions.

$8!$

- b) If each couple is to sit together.

$4!$ Ways for couples to sit and in each couple a member can be interchanged. Since 4 couples so interchanging will be 2^4 . Therefore $(2^4) \times (4!) = 384$ ways.

- c) If all the men sit together to the right of all the women.

$4!$ Ways for men to sit and $4!$ For women to sit. $4! \times 4! = 576$ ways. (If direction of right wasn't specified then we multiply by 2 for both left and right cases).

Q39) 8 finalists consist of 3 boys and 5 girls. Find number of sample points in the sample space S for number of possible orders at the conclusion of the contest for:

a) All 8 finalists

Any finalist can come in any order so $8!$

b) First 3 positions

$8 \times 7 \times 6 = 336$ or $8P3$.

Q40) In how many ways can 5 starting positions on a team can be filled with 8 men who can contribute at any positions?

$8P5 = 6720$. (That's because each position is different, if position means the same place then it becomes a combination problem).

Q41) Find the number of ways that 6 teachers can be assigned to 4 sections if no teacher is assigned to multiple sections?

$6P4$.

Q42) 3 lottery tickets for 1st, 2nd, 3rd round are drawn from a group of 40 tickets. Find the number of sample points in S for awarding the 3 prizes, considering each contestant holds only 1 ticket

First ticket can be won by any of 40 contestants, second by 39 and third by 38. So $40 \times 39 \times 38 = 59280$. Or $40P3$.

Q43) In how many ways can 5 different trees be planted in a circle

Since the formula for circular arrangement is $(n-1)!$. It's $4!$

Q44) Same

Q45) How many distinct permutations can be made from the letters of the word INFINITY?

Total letters = 8. I=3, N=2, FNY=1. So $8! / 3! \times 2! \times 1! \times 1! \times 1! = 3360$.

Q46) In how many ways can 3 oaks, 4 pines, 2 maples can be arranged along a line if one does not distinguish among the trees of the same kind?

$9! / 3! \times 4! \times 2! = 1260$.

Q47) How many ways are there to select 3 candidates from 8 graduates for job opening?

This is a combination problem because we are choosing 3 candidates for 1 position from 8 people so $r=3$, $n=8$. ${}^8C_3 = 56$.

Q48) How many ways are there that no two students will have the same birth date in a class of size 60?

Each student has a birthday from one of 365 days. For 60 students we are choosing 60 days from 365. Therefore 60 C 365.

PROBABILITY QUESTIONS SOLVED (2.49-2.65)

Q49) Find errors in each of the following statements:

- a) The probability that an automobile salesperson will sell 0,1,2,3 cars on any given day is 0.19,0.38,0.29,0.15.
Since $0.19+0.38+0.29+0.15=1.01$. This exceeds 1 which means it's false.
- b) The probability that it will rain tomorrow is 0.40 and the probability that it will not rain tomorrow is 0.52
 $0.40+0.52=0.92$ So it should not be less than 1.
- c) The probabilities that a printer will make 0, 1, 2, 3, or 4 or more mistakes in setting a document are, respectively, 0.19,0.34,-0.25,0.43, and 0.29.
Probability can't be negative
- d) The probability of selecting a heart is $\frac{1}{4}$ and the probability of selecting a black card is $\frac{1}{2}$ and the probability of selecting both heart and a black card is $\frac{1}{8}$.
In cards, hearts and diamonds are red and spades and clubs are black. It's impossible that a selected card is both black and a heart.

Q51) A box contains 500 envelopes of which 75 contains 100\$ in cash, 150 contain 25\$ and 275 contain 10\$. An envelope can be purchased for 25\$. What is the sample space for the different amounts of money? Assign probabilities to the sample points and find the probability that the first envelope purchased contains less than 100\$.

SS = {10\$,25\$,100\$} with weights: $275/500$, $150/500$, $75/500$.

If $P(10\$) = 275/500$ and $P(25\$) = 150/500$ The probability that the selected envelope contains less than 100\$ is the union of both i.e $275/500+150/500 = 0.85$.

Q52) In a class of 500 students, 210 students smoke, 258 drink, 216 eat between meals. 122 smoke and drink, 83 eat between meals and drink, 97 smoke and eat between meals. 52 students engage in all 3 of these bad practices. If a member of this class is selected, find the probability that the student:

S=210, D=258, E=216, SND = 122, DNE=83, SNE=97, SNDNE=52

a) Smoke but does not drink alcohol

Since smokers are 210, and smokers who drink are 122. We subtract them: $S - SND = 210 - 122 = 88$. So probability is $88/500 = 0.176$.

b) Eat between meals and drink alcohol but does not smoke

Since $DNE = 83$, we must subtract those who smoke so $DNE - SNDNE = 83 - 52 = 31$. Now probability is $31/500$.

c) Neither smokes nor eats between meals.

Smokers=218 and Eaters=216, and those who either smoke or eat = $SUE = S + U - SNE = 329$. Now we want to find those who DON'T do this so complement: $1 - SNE = 500 - 329 = 171$. Probability is $171/500 = 0.342$.

Q53) The probability that an American industry located in Shanghai is 0.7 and the probability that it will locate in Beijing is 0.4 and the probability that it will locate in either Beijing or Shanghai is 0.8. Find the probability that the industry will locate in:

a) In both cities

$$P(SNB) = P(S) + P(B) - P(SUB) = 0.7 + 0.4 - 0.8 = 0.3$$

b) In neither cities

$$P(S'NB') = 1 - P(SUB) = 1 - 0.8 = 0.2$$

Q54) The probability of a customer investing in tax-free bonds is 0.6 and the probability that it will invest in mutual funds is 0.3. The probability of the customer investing in both is 0.15. Find the probability that the customer will invest in:

a) Either tax-free bonds or mutual funds

$$P(TUM) = P(T) + P(M) - P(TNM) = 0.6 + 0.3 - 0.15 = 0.75.$$

b) Neither tax-free bonds not mutual funds

$$P(T'NB') = 1 - P(TUM) = 1 - 0.75 = 0.25$$

Q55) If each coded item in the catalog begins with 3 distinct letters followed by 4 distinct non-zero digits. Find the probability of randomly selecting one of these coded items with the first letter a vowel and the last digit an even.

Total possible combinations are $(26 \times 25 \times 24) \times (9 \times 8 \times 7 \times 6) = 47174400$.

Combinations of the first letter being vowel and last digit being even are $(5 \times 25 \times 24) \times (8 \times 7 \times 6 \times 4) = 4032000$.

Hence probability is $4032000/47174400 = 0.085$.

Q56) In a car, there is a probability of 0.25 of a defect in a brake system, 0.18 of a defect in transmission and 0.17 of a defect in the fuel system and 0.40 of a defect in some other area.

- a) Find probability that the defect is the brakes or the fuel system if the probability of defects in both systems together is 0.15

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.25 + 0.17 - 0.15 = 0.27$$

- b) Find probability that there are no defects in either brakes or fuel system

$$P(A' \cap B') = 1 - P(A \cup B) = 1 - 0.27 = 0.73$$

Q57) If a letter is chosen at random from the English alphabet. Find the probability that the letter is:

- a) Vowel, exclusive of y

$$5/26 = 0.192$$

- b) Listen somewhere ahead of letter j

Ahead means letters who come before which are 9. So $9/26 = 0.346$

- c) Listed somewhere after letter g

$$19/26 = 0.731$$

Q58) A pair of dice is tossed, Find the probability of getting:

- a) A total of 8

$$SS = \{(6,2), (2,6), (3,5), (5,3), (4,4)\} \quad .5/36 = 0.139$$

- b) At most a total of 5

$$SS = \{(1,1)(1,2)(2,1)(2,2)(2,3)(3,2)(4,1)(1,4)(3,1)(1,3)\}$$

Q59) In a poker hand consisting of 5 cards, find the probability of holding:

- a) 3 aces

Total ways to choose 5 cards from a deck of 52 is $52C5 = 2598960$

Total ways to choose 3 aces from available 4 is $4C3 = 4$

For 5 cards, we need to choose 2 more that are not ace, they can be any from the remaining 48 cards and the ways of doing so is $48C2 = 1128$

Hence ways to pick 3 ace in hand of 5 is $4 \times 1128 = 4512$

And probability of getting it is $4512/2598960 = 0.002$

- b) 4 hearts and 1 club

Total ways to choose 5 cards from a deck of 52 is $52C5 = 2598960$

Total ways to choose 4 hearts from available 13 is $13C4 = 715$

Total ways to choose 1 club from available 13 is $13C1 = 13$

Total ways of doing both is $13 \times 715 = 9295$

Probability of getting it is $9295/2598960 = 0.004$

Q60) If 3 books are picked at random from a shelf containing 5 novels, 3 poems, 1 dictionary. What is the probability of selecting:

- a) Dictionary

$9C3 = 84$ total ways to choose 9 books from 9

If 1 dictionary is picked, other 2 can be random so $8C2 = 28$

Hence Probability is $(24 \times 1)/84 = 0.286$

b) 2 novels and 1 poem

Total ways = ${}^9C_3 = 84$

To select 2 novels: ${}^5C_2 = 10$

To select 1 poem: ${}^3C_1 = 3$

Ways to do both = $10 \times 3 = 30$ and Probability of doing so is $30/84 = 0.357$

Q61) In a class of 100 students, 54 study maths, 69 study history, 35 study both maths and history. If one student is selected randomly. Find probability that the student took:

a) Maths or History

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 54/100 + 69/100 - 35/100 = 0.88$$

b) Did not take either of these subject

$$P(A' \cap B') = 1 - 0.88 = 0.12$$

c) Took history but not math

History students are 69 but from them some study both so subtract: $69 - 35 = 34$.

Hence $34/100 = 0.34$

Q62) In the pizza industry, consider a study involving 3 types of crusts and sauces.

a) How many combinations of crust and sauce are involved

3 crust and 3 sauces to total possible combinations = $3 \times 3 = 9$

b) What is the probability that a judge will get a plain thin crust (1 of type) with a standard sauce (1 of type) for his first taste test

It's one of the 9 combinations so $1/9$

Q63) The probable location of PCs in the home are as follows: Adult bedroom=0.03, Child bedroom=0.15, Other bedroom=0.14, Office=0.40, Other rooms = 0.28

a) What is the probability of a PC being in a bedroom?

$$P(A \cup B \cup C) = 0.03 + 0.15 + 0.14 = 0.32$$

b) What is the probability of a PC not being in a bedroom?

$$P(A \cup B \cup C)' = 1 - 0.32 = 0.68$$

c) In what room would you expect a PC the most if a random house is selected?

The office

Q64) Probability of an electric component surviving for more than 6000 hours is 0.42. Suppose that the probability that the component services no longer than 4000 hours is 0.04.

a) What is the probability that the life of the component is less than or equal to 6000 hours?

More than 6000 hours has probability=0.42. So not more than 6000 hours will have $1 - 0.42 = 0.58$ probability

b) What is the probability that life is greater than 4000 hours?

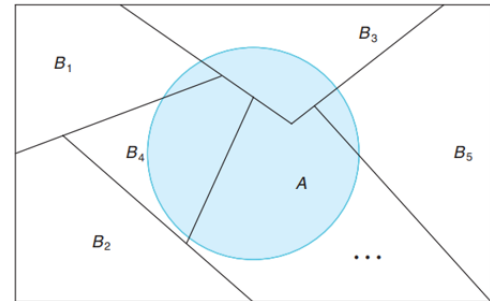
Life no longer than 4000 hours is 0.04 so the probability of having the life greater than 4000 hours is $1 - 0.04 = 0.96$

TOTAL PROBABILITY AND BAYES THEOREM

TOTAL PROBABILITY

If the events $B_1, B_2, \dots, B_k$ constitute a partition of sample space S such that $P(B_i) \neq 0$ for $i=1, 2, \dots, k$. Then for any event A of S :

$$P(A) = \sum_{i=1}^k P(B_i \cap A) = \sum_{i=1}^k P(B_i)P(A|B_i).$$

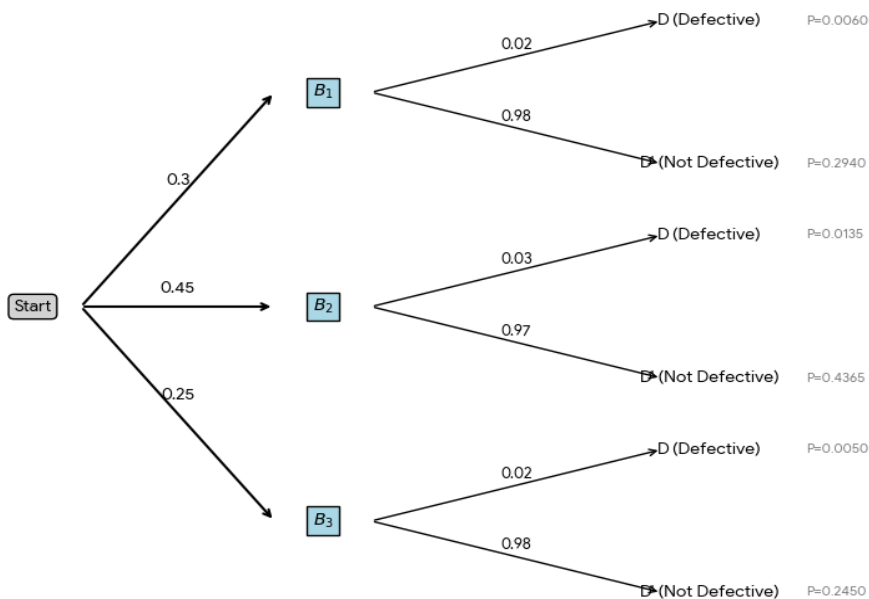


Q) In a certain assembly plant, three machines B_1, B_2, B_3 make 30%, 45%, 25% respectively of the products. It is known that 2%, 3%, 2% of products are defective. Now if a finished product is randomly selected, what is the probability of it being defective?

These types of questions are easily solved by a tree diagram. Given is:

$P(B_1)=0.3$, $P(B_2)=0.45$, $P(B_3)=0.25$ and $P(D|B_1)=0.02$, $P(D|B_2)=0.03$, $P(D|B_3)=0.02$

Probability Tree Diagram for Machine Defects



We have drawn a tree diagram of every possibility of the product. Now for defective. We have 3 paths so we will multiply the values on each path and then sum them. So the probability of the product selected being defective and being from B_1 is $(0.3)(0.02)=0.006$, from B_2 is $(0.45)(0.03)=0.0135$ and from B_3 is $(0.25)(0.02)=0.005$. Now the total probability that the product is defective is $0.006+0.0135+0.005=0.0245$.

BAYER'S RULE

If the events $B_1, B_2, \dots, B_k$ constitute a partition of the sample space S such that $P(B_i) \neq 0$ for $i=1, 2, \dots, k$. Then for any event A in S such that $P(A) \neq 0$:

$$P(B_r|A) = \frac{P(B_r \cap A)}{\sum_{i=1}^k P(B_i \cap A)} = \frac{P(B_r)P(A|B_r)}{\sum_{i=1}^k P(B_i)P(A|B_i)} \quad \text{for } r = 1, 2, \dots, k.$$

Q) Consider the previous Question, now if a defective product is selected, what is the probability that it came from machine B3?

For this we will take the branch "Machine B3 and Defective) i.e $P(D|B_3)=0.005$ and divide by Total Probability, because this is the rule: so $0.005/0.245=10/49$.

Q) A firm uses analytical plans to design costs of a product. Plans 1,2,3 are used for 30,20,50 percent of the products and the defective rate of each product is: $P(D|P_1)=0.01$, $P(D|P_2)=0.03$, $P(D|P_3)=0.02$. Where $P(D|P_j)$ is the probability that the product is defective, given plan j . Now if a randomly selected product is defective, which plan was the most likely responsible?

Since the total probability is: $(0.3)(0.01)+(0.2)(0.03)+(0.5)(0.02)=0.019$.

To find the most likely one, calculate for all and then find the result. We can do this by: Probability that Plan j was responsible = Defective Path from Plan j / Total Probability

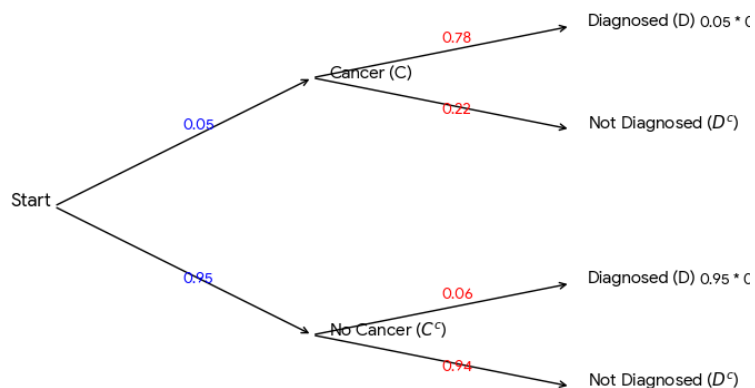
$$P(P_1|D) = (0.01)(0.3)/0.019 = 0.158$$

$$P(P_2|D) = (0.03)(0.2)/0.019 = 0.316$$

$$P(P_3|D) = (0.02)(0.5)/0.019 = 0.526$$

Since Plan 3 has the highest probability, it was the most likely responsible.

Q) The probability of selecting an adult over the age of 40 with cancer is 0.05. If the probability of the doctor correctly diagnosing cancer in a patient is 0.78 and incorrectly diagnoses cancer in an adult who is not a patient is 0.06. What is the probability that 40+ year old adult is diagnosed having a cancer.



Means we need
 Start → Cancer → Diagnosed +
 Start → No Cancer → Diagnosed
 Which is $0.05 \times 0.78 = 0.039$ plus
 $0.95 \times 0.06 = 0.057$.

$$\text{Ans: } 0.039 + 0.057 = 0.096$$

Q) Now what is the Probability a person actually has cancer

given that the doctor diagnosed cancer.

Doctor diagnosed cancer so total probability is $P(D)=0.096$

Now Acc to Bayer's Theorem: $P(C|D) = P(C \cap D)/P(D)$

$P(C \cap D)$ means patient has cancer And is correctly diagnosed so $0.78 \times 0.05 = 0.039$

Therefore $P(C|D) = 0.039/0.096 = 0.406$.

Means 40.6% of Patients who were diagnosed are really cancerous.

CONDITIONAL PROBABILITY

It is the likelihood that event B will occur, given that event A had already occurred.

The conditional probability of B, given A, denoted by $P(B|A)$, is defined by

$$P(B|A) = \frac{P(A \cap B)}{P(A)}, \quad \text{provided } P(A) > 0.$$

Suppose a squared dice where the probability of an even number occurring is twice as odd. So if the probability for an odd number is w then for even its $2w$. Since total probability sums up to $w+2w+w+2w+w+2w=9w$ so $w=1/9$ (for odd) and $w=2/9$ (for even). Now suppose it's known that toss of dice results in a number greater than 3 so now sample space is $\{4,5,6\}$. Now the total probability sums up to $2w+w+2w=5w$ so now $w=1/5$ (for odd) and $w=2/5$ (for even). This means that events may have different probabilities when considered relatively with different sample sizes.

Consider this table, and find the probability that selected male is employed

	Employed	Unemployed	Total
Male	460	40	500
Female	140	260	400
Total	600	300	900

Given that the selected person is male, sample space is no 600 so $P(E|M) = 460/600 = 23/30$.

Q) The probability that a regularly scheduled flight departs on time is $P(D)=0.83$; the probability that it arrives on time is $P(A)=0.82$; and the probability that it departs and arrives on time is $P(D \cap A)=0.78$. Find the probability that a plane:

- Arrives on time, given that it departed on time
- Departs on time, given that it arrived on time

- Event A is dependent on event D so $P(D|A) = P(D \cap A)/P(A) = 0.78/0.82 = 0.95$
- Event D is dependent on event A so $P(D|A) = P(D \cap A)/P(A) = 0.78/0.82 = 0.95$

Now calculate probability $P(A|D')$ that is that plane arrives on time given that it did not depart on time.

$$P(D' \cap A)/P(D') = 0.82 - 0.78 / 1 - 0.83 = 0.24$$

(The reason for doing $0.82 - 0.78$ is that when we in compliment do $1 - P$ it means that from whole sample space we are subtracting but in $A \cap B$ it means sample space is event A so we subtract from it).

Q) In testing if a cloth is defective or not, it is found that the length of strips are defective is 10% and for failure of texture its 5%, and 0.8% for both. If a cloth is selected randomly, find the probability that it fails the texture test if it has already defective length.

$P(L) = \text{defective length} = 0.10$ and $P(L \cap T) = 0.008$.

So $P(T|L) = P(L \cap T)/P(L) = 0.008/0.10 = 0.08$

INDEPENDENCE

In conditional probability the events are dependent, but if A and B are independent events, then $P(A|B) = P(A)$ or $P(B|A) = P(B)$.

Suppose you have 52 cards and event A is drawing an ace card and event B is drawing a spade card. Event A occurred and ace is drawn with probability $4/52=0.077$. Now if the events are dependent then sample space is reduced so for event B we do $13/51$. But in independence if event A had no impact (i.e different stack of cards), then we would do $13/52$.

Two events A and B are only independent if and only if $P(A \cap B) = P(A) \cdot P(B)$

MULTIPLICATIVE/PRODUCT RULE

If in an experiment the events A and B can both occur, then

$$P(A \cap B) = P(A)P(B|A), \text{ provided } P(A) > 0.$$

(derived from

basic formula). Basically in these questions we keep multiplying the probabilities from 1st choice.

Q) Consider a box of 20 fuses out of which 5 are defective. If 2 fuses are selected and removed from the box in succession (consecutively) without replacing the first. What is the probability that both fuses are defective.

To select the first fuse: $5/20 = 1/4$.

Now since its not replaced, selecting the second fuse means: $4/19$.

Hence total probability that both fuses are defective is $P(A \cap B) = 1/4 \times 4/19 = 1/19$.

Q) A bag contains 4 white and 3 black balls and another bag contains 3 white and 5 black balls. One ball is drawn from the first bag and placed in the second unseen. What is the probability that the ball drawn from second bag is black

When the ball is drawn from bag 1, if it is a white then its probability is $4/7$ and if its a black then its probability is $3/7$.

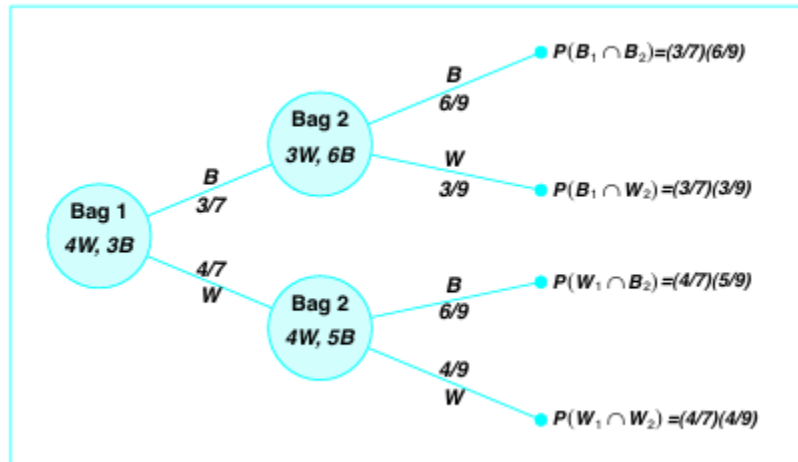
White Case: Now bag 2 has 4 white (3+1) and 5 black balls so the probability of drawing a black ball from it is $5/9$.

Hence the probability that the selected ball is black is $(4/7) \times (5/9) = 0.317$

Black Case: Now bag 2 has 3 white and 6 black balls (5+1)

so the probability of drawing a black ball from it is $6/9$.

Hence the probability that the selected ball is black is $(3/7) \times (6/9) = 0.286$



Now summing both case gives us $0.317 + 0.286 = 0.603 = 38/63$.

Q) A small town has one fire engine and one ambulance for emergencies. The probability that the fire engine is available when needed is 0.98 and the probability that the ambulance is available when needed is 0.92. Find the probability that both the ambulance and the fire engine are available together in a serious emergency situation, if both are independent.

Since both are independent: we know in this case $P(A \cap B) = P(A) \cdot P(B)$ so $0.98 \times 0.92 = 0.902$.

Q) Three cards are drawn in succession, without replacement, from an ordinary deck of playing cards. Find the probability that the event $A_1 \cap A_2 \cap A_3$ occurs, where A_1 is the event that the first card is a red ace, A_2 is the event that the second card is a 10 or a jack, and A_3 is the event that the third card is greater than 3 but less than 7.

$P(A_1) = 2/52, P(A_2) = 8/51, P(A_3) = 12/50$.

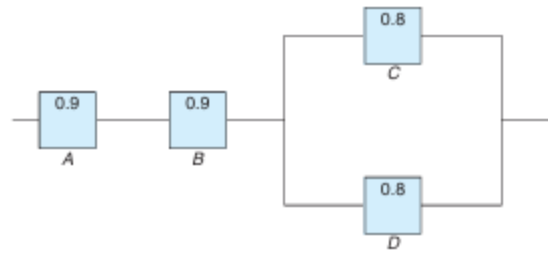
Hence $P(A_1 \cap A_2 \cap A_3) = (2/52) \times (8/51) \times (12/50) = 0.001$

Hence if a collection of events $A = \{A_1, \dots, A_n\}$ are mutually independent if for any subset of A , $A_{i1}, \dots, A_{ik}$, for $k \leq n$, we have $P(A_{i1} \cap \dots \cap A_{ik}) = P(A_{i1}) \dots P(A_{ik})$.

Q) An electrical system works as follows. Find the probability that:

- a) Entire system works
- b) Component C does not work, given that entire system does

The four components work independently.



- a) For the entire system to work, the probability can be calculated as $P(ANBN(CUD)) = P(A) \times P(B) \times P(CUD)$. To find $P(CUD) = P(C) + P(D) - P(CND)$. Since $P(CND) = P(C) \times P(D) = 0.8 \times 0.8 = 0.64$. So $P(CUD) = 0.8 + 0.8 - 0.64 = 0.96$.
Hence Answer = $(0.9 \times 0.9) \times (0.96) = 0.778$
- b) It is conditional probability case where event above is true but the given one is not so $P = P(ANBNC'ND) / P(\text{system})$
 $P(A) \times P(B) \times (1 - P(C)) \times P(D) = 0.9 \times 0.9 \times (1 - 0.8) \times 0.8 = 0.1296$
So $P = 0.1296 / 0.778 = 0.1667$